

WEEK – 03

A) Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.

```
#include <stdio.h>
#include <time.h>

#define MAX 100
#define INF 9999

int main() {
    int n, cost[MAX][MAX];
    int visited[MAX] = {0};
    int i, j, edges = 0;
    int u, v, min;
    int min_cost = 0;

    clock_t start, end;
    double time_taken;

    printf("Enter number of vertices: ");
    scanf("%d", &n);

    printf("Enter adjacency matrix:\n");
    for(i = 0; i < n; i++) {
        for(j = 0; j < n; j++) {
            scanf("%d", &cost[i][j]);
            if(cost[i][j] == 0)
                cost[i][j] = INF;
        }
    }

    start = clock();

    visited[0] = 1;

    printf("\nEdges in MST:\n");

    while(edges < n - 1) {
        min = INF;
        u = v = -1;

        for(i = 0; i < n; i++) {
```

```

        if(visited[i]) {
            for(j = 0; j < n; j++) {
                if(!visited[j] && cost[i][j] < min) {
                    min = cost[i][j];
                    u = i;
                    v = j;
                }
            }
        }
    }

    printf("%d - %d : %d\n", u, v, min);

    min_cost += min;
    visited[v] = 1;
    edges++;
}

end = clock();

time_taken = ((double)(end - start)) / CLOCKS_PER_SEC;

printf("\nTotal cost of MST = %d\n", min_cost);
printf("Time taken = %f seconds\n", time_taken);

return 0;
}

```

OUTPUT:

```

PS C:\Users\SHALINI K A\Desktop\ADA> ./Prims
Enter number of vertices: 4
Enter adjacency matrix:
0 1 2 3
0 3 0 1
0 5 2 1
1 2 3 4

Edges in MST:
0 - 1 : 1
1 - 3 : 1
0 - 2 : 2

Total cost of MST = 4

```

B) Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.

```
#include <stdio.h>
#include <time.h>

#define MAX 100
#define INF 9999

int parent[MAX];

int find(int i) {
    while(parent[i] != i)
        i = parent[i];
    return i;
}

void union_sets(int a, int b) {
    parent[b] = a;
}

int main() {
    int n, e;
    int u[MAX], v[MAX], w[MAX];
    int i, j;
    int min_cost = 0;

    clock_t start, end;
    double time_taken;

    printf("Enter number of vertices: ");
    scanf("%d", &n);

    printf("Enter number of edges: ");
    scanf("%d", &e);

    printf("Enter edges (u v w):\n");
    for(i = 0; i < e; i++) {
        scanf("%d %d %d", &u[i], &v[i], &w[i]);
    }

    for(i = 0; i < n; i++)
        parent[i] = i;
```

```

start = clock();

printf("\nEdges in MST:\n");

for(i = 0; i < e; i++) {
    int min = INF, index = -1;

    // find minimum edge
    for(j = 0; j < e; j++) {
        if(w[j] < min) {
            min = w[j];
            index = j;
        }
    }

    w[index] = INF;

    int a = find(u[index]);
    int b = find(v[index]);

    if(a != b) {
        printf("%d - %d : %d\n", u[index], v[index], min);
        min_cost += min;
        union_sets(a, b);
    }
}

end = clock();

time_taken = ((double)(end - start)) / CLOCKS_PER_SEC;

printf("\nTotal cost = %d\n", min_cost);
printf("Time taken = %f seconds\n", time_taken);

return 0;
}

```

OUTPUT:

```
● PS C:\Users\SHALINI K A\Desktop\ADA> gcc Kruskals.c -o Kruskals
● PS C:\Users\SHALINI K A\Desktop\ADA> ./Kruskals
Enter number of vertices: 4
Enter number of edges: 5
Enter edges (u v w):
0 1 10
0 2 6
0 3 5
1 3 5
2 3 4

Edges in MST:
2 - 3 : 4
0 - 3 : 5
1 - 3 : 5

Total cost = 14
```